Hidden States Turn Chain-of-Thought Tokens into Working Memory

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Abstract

Where do language models store the intermediate values used in multi-step reasoning? One hypothesis is that models reason in an internal workspace and write chain of thought only when that workspace becomes insufficient. We find instead that chain-of-thought tokens function as **token-slot memory**: the hidden state at each written value acts as a slot that stores the value, is read by later computation, and can be set independently of the visible text. Holding a model's reasoning text fixed to the character, we overwrite the hidden state at one written number with the state from a run that wrote a different number: the final answer follows the never-written number in 70-100% of runs on six off-the-shelf models from four families (4B to 14B), size-matched random controls reach at most 1% on the four models without reasoning distillation, and on Qwen3-4B the planted state stays causally effective across 40 subsequent reasoning steps. Later computation retrieves the stored value by attending to its position (blocking that attention removes the effect), and two positions hold independently settable values that compose into a sum appearing nowhere in the text (88-90% of runs). The capacity-triggered account finds no support: forbidden to write, models from 1.5B to 671B parameters collapse beyond one or two dependent steps, yet writing shows no onset threshold (93-100% of intermediate values written from depth one), and no intervention, including ablating the selected top-16 concept directions of J-space, a recently proposed residual-stream workspace, moves state between the trace and internal stores. For oversight, an identical visible reasoning prefix can carry different hidden states that lead to different answers; whether such divergences arise without intervention is open.

1 Introduction

A language model doing multi-step reasoning computes a value, writes it into its chain of thought, and uses it several steps later. The model could maintain the value independently of its output, or bind it to the emitted token position and retrieve it through attention. For serial numeric reasoning we give a causal answer: the state is bound to the written position. We call this token-slot memory: each written value's token is a slot, the hidden state at that position is the slot's content, and later computation reads the content by attending to the position. It qualifies as working memory operationally: the content persists at its slot, is selectively retrieved during later computation, and can be manipulated independently across slots.

A reasoning prefix ends with the line "After step 5 the value is 457." (Figure 1A). Every visible character stays in place; we overwrite the hidden state at the 457 position with the state from a run that wrote 462 there, and the final answer builds on 462, a number appearing nowhere in any text. Among items where the model demonstrably relies on the written intermediate, the planted

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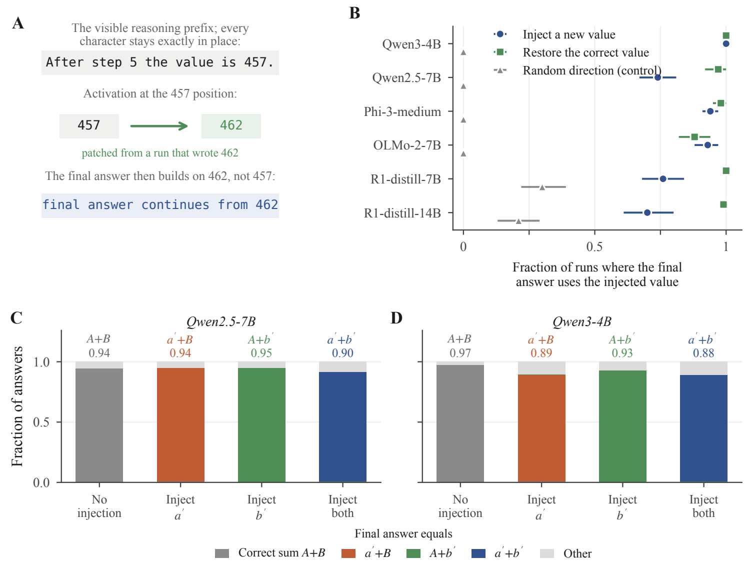

Figure 1: Same visible prefix, different stored value, different answer. (A) The reasoning prefix ends "After step 5 the value is 457." The text never changes; the activation at the 457 position is replaced with one captured from a run that wrote 462, and the final answer continues from 462, a value appearing nowhere in any text. (B) Fraction of runs whose final answer uses the injected value, per model: planting a counterfactual value p (circles), restoring the correct value c under edited text (squares), and a size-matched random control (triangles); bars are 95% bootstrap intervals (the distilled models' elevated random control is explained in Appendix A). (C, D) With two counters A and B, injecting a replacement for either or both (a', b') makes the answer equal the matching sum; answers using only one of two injected values occur at 0.01.

value redirects the final answer in 0.70 to 1.00 of runs across six off-the-shelf models from four families; size-matched random noise redirects at most 0.01 (Section 4). On Qwen3-4B the planted state remains causally effective across 40 subsequent reasoning steps. Section 3 shows writing is not capacity-triggered at inference time (behavioral, 1.5B to 671B); Section 4 that the hidden state at a written value is causally active and settable (causal, 4B to 14B); Section 5 that later computation retrieves it by attending to its position, in the middle third of layers; and Section 6 that two positions hold values that compose.

The contrasting account centers on J-space, a proposed global workspace identified by a Jacobian-based lens over residual activations [Gurnee et al., 2026], and it makes two separable predictions: a capacity-triggered workspace predicts writing begins at the depth where silent performance fails, and a causally load-bearing workspace predicts that removing its directions impairs reasoning. Section 3 rejects both: writing begins at step one at every scale, and ablating the selected top-16 J-space concept directions has no greater effect than ablating size-matched random directions at any tested strength. Causally important task state need not be localized to the concept directions a fitted lens selects.

Prior work anticipated the memory role of written tokens theoretically [Merrill and Sabharwal, 2024], behaviorally [Lanchantin et al., 2023], and causally on a fine-tuned model [Shih et al., 2026] and on program-variable tasks [Zhu et al., 2025]; the contribution here is a causal account of storage, retrieval, and composition on off-the-shelf models.

2 Experimental setup

Both task families give exact ground truth for every intermediate value, so the downstream effect of any edit is computable. Arithmetic chains start from a three-digit number and apply d signed two-digit additions and subtractions. Narrative state-tracking problems describe a character whose count of an item changes through varied events and surface forms. In both, the model writes one line

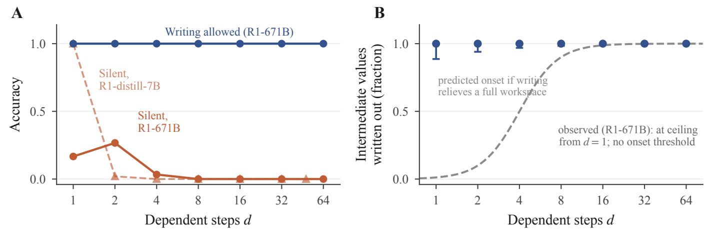

Figure 2: Writing is not triggered by exhausted internal capacity. (A) Instructed to answer without writing, models collapse after one to two dependent steps; with writing allowed, accuracy stays near ceiling (n=30 items per depth). (B) The fraction of intermediate values written is 0.93 to 1.00 at every depth (Wilson intervals); the dashed curve is the onset a capacity-relief account predicts, and it does not appear.

per step giving the running value. A third task, box tracking, updates several containers in parallel; it appears only as the internally held comparison for the lesion test. White-box experiments use Qwen3-4B, Qwen2.5-7B-Instruct, Phi-3-medium, OLMo-2-7B-Instruct, and DeepSeek-R1-Distill-Qwen 7B and 14B; the 1.5B distill appears in the behavioral scale range only. Frontier models (DeepSeek V3.2, DeepSeek-R1-671B, Claude Sonnet 4.5, Llama 3.3 70B) are tested behaviorally through assistant prefill.

The protocol takes a correct worked solution, edits the value at one step, truncates immediately after the edited line, and lets the model continue. Three values recur: the correct c from the original solution, the edited e shown in the text, and a counterfactual p whose activation is planted. The conditions: a *text edit* shows e with no state intervention; *restore* shows e with the position's hidden state restored to c's; *plant* shows e with the state replaced by p's, captured from a run that wrote p; the *random control* shows e with a size-matched random perturbation. The patched state is the residual-stream activation at the value's position across a middle band of layers; interventions cover every token of multi-token values (Appendix C). A no-edit truncation measures the sampling noise floor, 0.00 throughout. Generation uses temperature 0.6 and top-p 0.95 with five samples per item; unparseable outputs are counted separately. Intervals are 95% bootstrap intervals over items.

3 Write: no inference-time capacity threshold

Instructed to output only the final answer (Figure 2A), with token counts confirming nothing else was generated, models fail beyond one to two dependent steps (Qwen2.5-7B: 1.00 at $d = 1$, 0.04 at $d = 4$; DeepSeek-R1-671B: at most 0.27 at any depth). Inert filler tokens in place of a trace [Pfau et al., 2024] do not rescue the collapse (Appendix C). The values are random, so nothing can be recalled; the failures bound performance under this instruction, with its prompting and training-distribution effects, rather than latent internal capacity.

Models write 0.93 to 1.00 of ground-truth intermediates in free generation from $d = 1$ through $d = 64$, at every scale tested: no difficulty threshold at which writing begins is observed. This rules out an inference-time switch from internal holding to writing; an always-write policy learned in training could itself descend from capacity limits.

Three interventions test whether state can move between internal and written stores. A residual-stream lesion at fixed dose lowers accuracy on box tracking, whose state is held internally, by 0.34, and on the written chain task by 0.11; a lesion strong enough to cross the accuracy cliff makes traces disintegrate rather than expand. Hard token budgets compress wording up to 2.5 times while keeping 0.98 to 1.00 of values, then truncate below roughly twelve tokens per step. Ablating the selected top-16 J-space concept directions at every position has no greater effect than ablating size-matched random directions: at a calibrated generation-preserving dose it leaves chain-of-thought accuracy, bare-answer accuracy, and the amount written unchanged on synthetic chains and on GSM8K, and accuracy on one- and two-step chains stays at ceiling at every dose up to complete removal of the selected directions (Appendix C). The chain-of-thought-versus-direct ablation asymmetry reported for J-space [Gurnee et al., 2026] therefore does not replicate on our tasks at any tested strength,

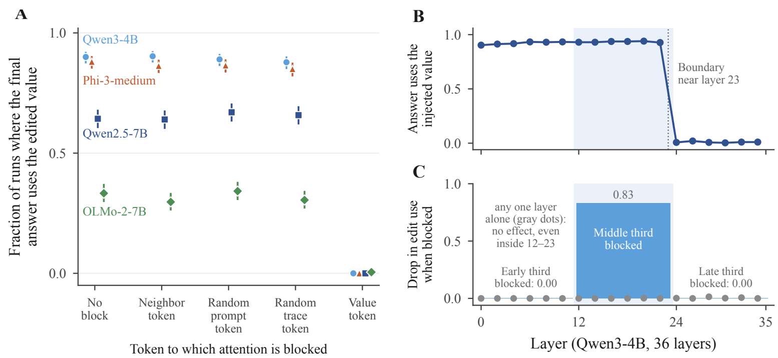

Figure 3: Written values are retrieved through attention to their token, across the middle third of layers. (A) Blocking later positions' attention to the edited value's token drops edit use to near zero on four models; matched control blocks change nothing. Points are means with 95% Wilson intervals (n=600 runs per condition). (B) Overwriting the hidden state at any single layer before the read redirects the answer; after it, nothing (Qwen3-4B). (C) Blocking the middle third removes 0.83 of edit use; the early and late thirds, and every single layer (gray dots), remove none.

including the depths where models do perform the arithmetic internally. Section 4 shows a causally decisive planted value sitting in the same residual states the lens reads.

4 Store: written positions hold causally active values

Figure 1 decomposes the effect of editing a written value, conditioned on the cases where the memory is demonstrably in use: items on which the model follows the visible text edit to e (0.44 to 0.98 by model; Appendix A). Among these, restoring c's state while the text still shows e returns the answer to correct in 0.88 to 1.00 of cases, and planting p's state redirects the answer to p, a value appearing nowhere in any text, in 0.70 to 1.00 of cases. The size-matched random control sits at or below 0.01 for the four models without reasoning distillation. The patch replaces the position's full contextual activation; it shows that state is causally sufficient to make later computation behave as if p had been written, and it does not isolate a numeric feature within that state. Because p is a mid-chain intermediate the model never produced, the patch does not inject the answer. The effect persists to 40-step chains (0.95 to 0.98 on Qwen3-4B) and replicates on the narrative task (Appendix A).

5 Retrieve: attention to the written position

Blocking attention from all post-edit positions to the edited value's position collapses the answer's dependence on that value on every model tested (Figure 3A); matched blocks of the adjacent words, a random prompt position, or a random trace position leave it at the unblocked level. Blocking the operand the next step needs breaks the arithmetic itself, a positive control for the mask. The collapse holds across three families and both task formats (natural-language: 0.730 to 0.007, 0.720 to 0.003; coherence control in Appendix B).

Layer-resolved versions of both interventions locate the read in depth on Qwen3-4B (36 layers; Figure 3B, C). Overwriting the value's hidden state at any single layer up to 22 redirects the answer at 0.90 to 0.94; at layer 24 and beyond, at 0.02 or less ($n = 60$): the value can be rewritten at any layer before the read and at none after. Blocking the value-position attention only at the middle third of layers (12 through 23) collapses following from 0.857 to 0.025, blocking only the early or late third leaves it unchanged, and no single layer's block matters ($n = 80$): the read is distributed redundantly across the middle third and confined to it; the two sweeps agree on the same boundary, near layer 23. A coarse sweep on Phi-3-medium shows the same structure; Qwen2.5-7B is unresolved at this granularity (Appendix B).

6 Compose: slots are independent

In a task with two counters summed at the end (Figure 1C, D), planting a never-written value at one final position yields the corresponding mixed sum (0.94 and 0.95 on Qwen2.5-7B, 0.89 and 0.93 on Qwen3-4B). Planting both yields their joint sum at 0.90 and 0.88 (products of the single rates: 0.89 and 0.83). The random patch leaves the clean answer in place (0.96 and 0.95). The reported number must be assembled at answer time from two slots set independently, holding values never written anywhere.

7 When models retrieve versus recompute

By default, models reuse the value they wrote rather than recompute it, and what triggers recomputation is model-specific: DeepSeek V3.2 reverts an edit when one sentence in the prompt states the value's derivation (0.43 following, against 0.93 when the prompt only implies the value), Claude Sonnet 4.5 when the edited step is the value's only source (0.22 against 0.81), while Llama 3.3 70B follows edits at 0.97 despite solving the same chains from scratch, and the Qwen models' following rates in Table 1 leave little room for reversion. The conditions are separate experiments and we claim within-experiment contrasts only (details in Appendix C). On GSM8K worked solutions, an edited intermediate is followed at 0.87 by Qwen3-4B and at 0.10 by V3.2, and within one model the contrast holds, V3.2's 0.93 on chains against 0.10 on GSM8K: a benchmark intermediate is often one operation from numbers still in the prompt. Lanham et al. [2023] report reliance on written reasoning declining with scale on standard benchmarks; our contrasts offer a candidate factor, the regenerability of the state, which influences reliance without determining it, and neither scale nor recomputation ability alone predicts it. Trace reliance is separate from verbalization faithfulness, whether the trace reports the computation behind the answer; our claims concern reliance on already-written state.

8 Related work

Our interventions build on causal tracing and activation patching [Meng et al., 2022] and causal abstraction [Geiger et al., 2021]; Heimersheim and Nanda [2024] caution that broad patches conflate a component mattering with deleting information at a position, which the single-layer sweep in Section 5 addresses. Faithfulness studies [Turpin et al., 2023, Lanham et al., 2023] and filler-token results [Pfau et al., 2024] bound our claims to state the model does not regenerate internally. Wang [2026] corrupt surface traces only in aggregate and conclude reasoning is latent; our token-level edits show written state is causally decisive when the model cannot regenerate it. Korbak et al. [2025] argue chain-of-thought monitoring is a useful and fragile oversight channel; our results bound what such a monitor observes.

9 Limitations

Tasks are numeric and count state tracking, so our claims concern serial task state; code, plans, and richer multi-entity state are untested. The J-space null bounds the fitted readout and the selected active directions, on our tasks and doses: the full-strength sweep covers six middle layers of one model, and the lens's domain excludes attention and MLP interiors, the KV cache, and residual directions outside the fitted dictionary. The fine layer sweeps cover one model. White-box evidence is at 4B to 14B scale, frontier evidence behavioral. Conditioning on edit-following makes cross-model levels not directly comparable (Appendix A). A structured patch control planting a valid activation for another quantity would sharpen the value-content claim.

10 Conclusion

Written positions function as token-slot memory in the serial reasoning tasks we study, a working memory whose slots are the model's own emitted tokens: a slot's hidden state can be changed independently of the visible prefix, is retrieved through middle-layer attention, and composes with other slots to determine later answers. Ablating the selected top-16 J-space concept directions leaves all of this intact. A monitor that reads the completed trace sees the text; under intervention, identical visible prefixes lead to different subsequent reasoning and answers [Korbak et al., 2025]; whether comparable divergences arise naturally, and in richer reasoning than numbers, remains open.

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A Per-model edit and patch rates

Table 1 gives the per-model rates behind Figure 1B, with the text-edit-following rate that conditions the patch columns. Ten-step chains; brackets are 95% bootstrap intervals; n counts items where the text edit was followed. Unconditioned per-run outcomes are in the released data. The distill models re-solve the problem after their reasoning phase, which elevates their random-control reversion: in a text-logged sample, 24 of 24 reverting random-control continuations re-derive from the prompt's starting value after the reasoning phase, and none correct the value in place. Re-solving can only produce the correct answer, so the planted-value rates are unaffected. R1-distill-1.5B is excluded (12 usable items; random control exceeds the planted-value rate). On the narrative task the pattern replicates: a planted never-written count is followed at 0.83 and 0.86, restoring the correct state returns the answer at 0.89 and 0.94, and the random control is 0.00, on Qwen3-4B and Phi-3-medium ($n = 86, 89$).

Table 1: Text-edit following and state-patch outcomes by model.

Model	Text edit followed	Restore correct state	Plant never-written value	Random control	n
Qwen3-4B	0.98 [0.95, 1.00]	1.00 [1.00, 1.00]	1.00 [0.99, 1.00]	0.00 [0.00, 0.00]	106
Qwen2.5-7B	0.85 [0.80, 0.90]	0.97 [0.92, 1.00]	0.74 [0.67, 0.81]	0.00 [0.00, 0.01]	93
Phi-3-medium	0.94 [0.90, 0.98]	0.98 [0.95, 1.00]	0.94 [0.91, 0.97]	0.00 [0.00, 0.00]	102
OLMo-2-7B	0.85 [0.79, 0.91]	0.88 [0.82, 0.94]	0.93 [0.88, 0.97]	0.00 [0.00, 0.00]	100
R1-distill-7B	0.60 [0.53, 0.66]	1.00 [0.99, 1.00]	0.76 [0.68, 0.84]	0.30 [0.22, 0.39]	65
R1-distill-14B	0.44 [0.36, 0.52]	0.99 [0.98, 1.00]	0.70 [0.61, 0.80]	0.21 [0.13, 0.29]	46

B Coarse cross-model layer sweeps

The state patch applied to one third of layers at a time ($n = 60$ items per model; fraction of runs following the planted value). Qwen3-4B (36 layers): early 0.91, middle 0.93, late 0.00, all layers 0.92. Phi-3-medium (40 layers): early 0.89, middle 0.87, late 0.01, all layers 0.89. Both match the fine sweep's structure: the planted value takes effect at any band before the read and at none after. Qwen2.5-7B (28 layers) follows the planted value at 0.70 to 0.73 in every band, so its read boundary is not resolved at this granularity.

Under the attention block of Section 5, the model continues fluently, with answers matching neither the edited nor the correct value (unparseable rate 0.000; OLMo-2-7B 0.040, its unblocked level). Reading the written value is the preferred path, with reconstruction from earlier lines a model-specific fallback: with the value's characters replaced by unreadable dots and attention untouched, the Qwen models reconstruct the value from the previous line and recover the correct answer in 0.60 to 0.73 of cases; Phi-3-medium never does (0.00).

C Generation and intervention configuration

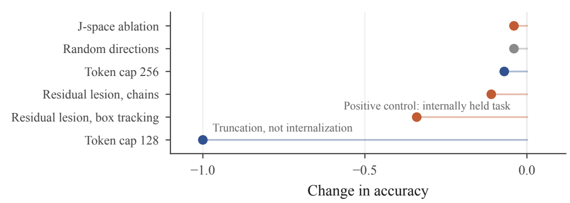

Figure 4: No intervention moves state between the workspace and the trace: J-space ablation and random directions leave accuracy unchanged; residual lesions hurt the internally held box-tracking task far more than the written chain task; a token cap below the per-step floor collapses accuracy by truncation.

Generation uses temperature 0.6, top-p 0.95, five samples per item. The text edit changes only the visible text of the value. A written value may span several tokens; the state patch and the attention block apply to every token of the value's span. The state patch overwrites the residual stream at those positions across a middle band of layers with activations captured from a run in which the counterfactual value was written there; the attention block masks all attention from post-edit positions to those positions at all layers. White-box sample sizes are 100 to 600 per condition with two to three seeds; the behavioral read-versus-recompute experiments of Section 7 contain 1,300 to 2,400 scored runs per cell.

The filler condition prefills the assistant turn with twelve inert dot tokens per dependent step, then decodes the answer at temperature 0. Full cells ($n = 30$ each): at $d = 2$, 0.97 with filler against 0.40 without on Qwen3-4B and 0.53 against 0.47 on Qwen2.5-7B; at $d = 4$, 0.07 against 0.00 and 0.13 against 0.07; at $d = 8$, 0.00 for both models in both conditions. Qwen3-4B's recovery at $d = 2$ is the parallel-width benefit filler tokens are known to provide, and it does not extend to the depths where writing is needed.

In the DeepSeek V3.2 evidence conditions of Section 7, with the derivation-stating sentence present but the edit two steps downstream, following returns to 0.96, so the reversion targets the stated value.

The J-space ablation dose is calibrated on neutral text to the strongest strength preserving ordinary generation (next-token agreement 0.87, perplexity ratio 1.12). The dose sweep runs the ablation of Section 3 at $k = 16$ and $\alpha \in \{0.125, 0.25, 0.5, 0.75, 1.0\}$ on layers 9 to 14 of Qwen2.5-7B with identical items across conditions; accuracy stays 0.99 to 1.00 in every condition on one- and two-step chains ($n = 80$ per cell). Prompts, model revisions, and per-run configuration files are released with the code and data.